

Assignment 1: Opinion Network Formation

Three Views of One Survey: Student–Student, Student–Question and
Question–Question Networks

Team **MATlab matlab?**

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September 2026

1 Team Name

MATlab matlab?

2 GitHub Repository

https://github.com/its-amish/DPCN_ass1

The repository is organised by network: **Student Network/** (Network A), **Extreme Bipartite Network/** (Networks B and C, plus interactive D3 visualisers), and **Question Network/** (Network D). The shared cleaning module `clean.py` sits at the repository root and is used by Networks A and D. Every figure and number in this report is produced by the notebook in the corresponding folder; the numbers quoted for Networks B, C and D were additionally re-derived from the raw CSV as a cross-check. This report is built from `report/report.tex`.

3 Dataset Documentation

The survey collects opinions on 60 statements, in four sections of 15: **T** Technology, **E** Education, **S** Society & Ethics, **V** Environment. There are **96 responses**, each answer on a five-point Likert scale, giving at most 5,760 data points.

3.1 What the raw answers look like

Answer	Count	Share of given answers
Strongly Agree	2,251	43.1%
Agree	1,914	36.7%
Neutral	686	13.1%
Disagree	256	4.9%
Strongly Disagree	111	2.1%
Blank	505	—
“No Comments”	37	—

Two features of this table drive every decision below. First, the class is **overwhelmingly agreeable**: 79.8% of all given answers are Agree or Strongly Agree and only 7.0% are any kind of disagreement. Second, **missing answers are not scattered**, they arrive in whole sections:

- 5 respondents (IDs 44, 60, 68, 73, 78) answered **nothing at all**;
- 4 respondents (IDs 30, 39, 77, 87) answered **only the Technology section**;
- ID 79 skipped all of Environment and ID 123 skipped 10 of its 15 questions;
- every other respondent left at most 4 answers blank.

Blank cells and the 37 “No Comments” answers are treated as missing throughout.

3.2 Two preparations of the same data

The three networks need different things from the data, so the project uses two documented preparations.

	Preparation 1 (Networks A, D)	Preparation 2 (Networks B, C)
Encoding	−2 Strongly Disagree ... +2 Strongly Agree; sign = polarity, size = strength	1 Strongly Disagree ... 5 Strongly Agree
Respondents removed	only the 5 completely empty responses → 91 students	respondents with no extreme answer → 89 students
Missing answers	kept as missing; each pair of students is compared only on questions <i>both</i> answered	dropped from the long-format edge list
Answers used	all five levels	only Strongly Disagree and Strongly Agree ($\{1, 5\}$)
Result	91 students $\times$ 60 questions, 242 missing cells	2,362 extreme responses

Preparation 1 is implemented once in `clean.py`, is shared by Networks A and D, and adds 3 to recover the conventional 1–5 scale, so the two encodings are interchangeable. One caveat should be recorded honestly: the 7 respondents that Preparation 2 drops for having “no extreme view” are mostly not opinion-moderates. Five of them submitted an empty form and one (ID 77) answered only Technology; **ID 119 is the only respondent who genuinely answered the survey without ever using an extreme**. Preparation 2’s remaining 89 students therefore still include three respondents (IDs 30, 39, 87) who answered only 15 of the 60 questions, and their node degrees in Networks B and C are correspondingly understated.

4 Pipeline Followed

We build four networks from the same survey, each answering a different question. Two look at the students, two at the questions, and the two question networks are built on deliberately different definitions of a link.

	Network A	Network B	Network C	Network D
Nodes	91 students	89 students + 60 questions	60 questions	60 questions
Edge means	“agree more than chance”	“holds an extreme view on”	“co-held extreme views”	“answers rise and fall together”
Type	weighted, one-mode	bipartite, unweighted	weighted projection of B	weighted, thresholded
Edges	2,026	2,362	1,770 (K_{60})	188
Author	Mayank Kar	Tanush Garg	Tanush Garg	Amish Goyal

Networks C and D are both question–question networks, but they are not competing answers to the same question. C asks *which questions provoke intense feeling in the same students*, and counts only extreme answers; D asks *which questions move together across the full five-point scale*, and uses every answer. Read together they separate intensity from alignment.

4.1 Network A — student–student agreement

Step 1: a simple agreement score. For every question both students answered, the pair scores 2 for an identical answer (including both Neutral), 1 for different answers on the same side of the scale (Agree/Strongly Agree, or Disagree/Strongly Disagree), and 0 otherwise. Averaging over the n_{ij} questions both answered and halving gives a score in $[0, 1]$:

$$w_{ij}^{\text{simple}} = \frac{1}{2n_{ij}} \sum_{q \in \text{shared}} s_q(i, j).$$

Averaging rather than summing matters because the Technology-only students share just 15 questions with everyone else.

Step 2: adjusting for chance. The simple score never reaches 0 (its minimum over 4,095 pairs is 0.133), so a network built from it links *every* student to every other. The cause is the answer distribution above: when four in five answers are on the agree side, two students drawn at random already match most of the time. We therefore subtract, for each question, the average score E_q over *all* pairs of classmates — what two randomly chosen students would score on that question:

$$w_{ij} = \frac{1}{n_{ij}} \sum_{q \in \text{shared}} (s_q(i, j) - E_q).$$

A positive weight now means “this pair agrees more than two random classmates would”. Matching on a question the whole class agrees on adds almost nothing; matching on a question that splits the class counts heavily. **Network A links every pair with $w_{ij} > 0$:** 2,026 edges, 49.5% of all pairs, one student isolated.

Step 3: sparsification. At 49.5% density the network is unreadable and its path lengths are uninformative, so we thin it two ways: a global cutoff swept from weak to strong (experiment 1), and keeping each student’s **two strongest edges** (experiment 2), which yields a 155-edge network used for drawing and for the sparse half of the measurements.

4.2 Network B — student–question extremes (bipartite)

Responses are melted into (student, question, response) triples and filtered to $r \in \{1, 5\}$, so an edge exists exactly when a student holds an *extreme* view on a question. Moderate and neutral answers are deliberately discarded: the aim is to map ideological intensity rather than consensus. The result is a verified bipartite graph $G = (U, V, E)$ with $|U| = 89$, $|V| = 60$, $|E| = 2,362$, bipartite density 0.4423, and a single connected component.

4.3 Network C — question–question projection

Network B is projected onto the question set:

$$e_{q_i, q_j} \in E_Q \iff \exists u \in U : (u, q_i) \in E \wedge (u, q_j) \in E,$$

i.e. two questions are linked when at least one student holds extreme views on both. The unweighted projection turns out to be the complete graph K_{60} , so all further analysis uses the **weighted** projection G_Q^W , where w_{ij} counts the students who hold extreme views on both questions.

4.4 Network D — question–question correlation

Network D starts from Preparation 1 (the same 91 students and the same $-2 \cdots + 2$ encoding as Network A) and asks, for each of the $\binom{60}{2} = 1,770$ question pairs, whether the two statements

are answered in step. The measure is the **Spearman rank correlation** ρ_{ij} , computed over the students who answered both. Spearman rather than Pearson because Likert answers are ordinal, so only the monotonic relationship is meaningful.

A correlation matrix is fully connected, so a cutoff is applied: an edge exists iff

$$\rho_{ij} > 0.35,$$

with weight $w_{ij} = \rho_{ij}$ and, for path-based measures, distance $d_{ij} = 1/\rho_{ij}$, so that strongly correlated statements lie close together. The cutoff discards weak and ambient associations and keeps moderate-to-strong ones. Eigenvector centrality then identifies the statements at the core of the belief structure and weighted betweenness the statements bridging otherwise separate groups. Two further tools describe the grouping. **Louvain** searches for a split of the questions that maximises modularity, the share of edge weight falling inside groups minus the share expected if the same edges were rewired at random; it is used here only to describe the clusters, not to make a claim about their strength. The **attribute assortativity coefficient** r is the correlation between the section labels (T, E, S, V) at the two ends of an edge: $r = 1$ would mean questions link only within their own section, $r = 0$ that the sections say nothing about which questions link.

5 Analysis and Visualizations

5.1 Network A: student–student agreement

Code and figures: Student Network/student_network.ipynb (Mayank Kar).

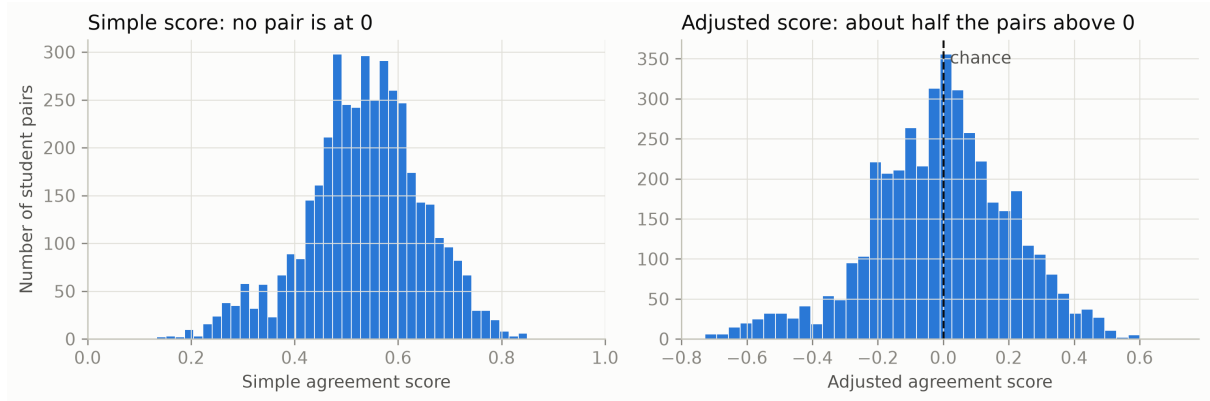


Figure 1: Left: the simple agreement score over all 4,095 pairs — no pair sits at 0, so this score cannot define a network. Right: after subtracting chance agreement the scores centre on 0, and “above chance” becomes a meaningful cutoff. 49.5% of pairs survive it.

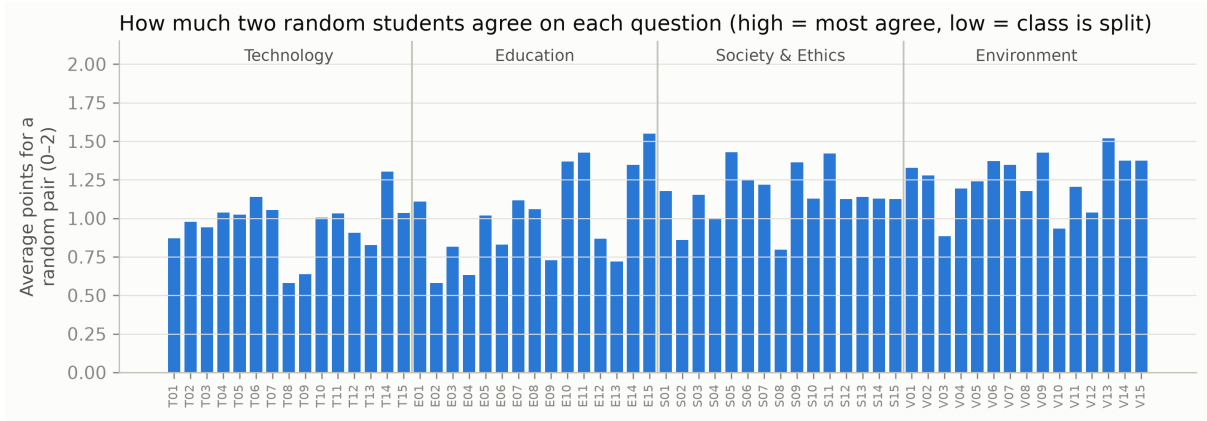


Figure 2: Chance agreement E_q per question. High bars are consensus questions, where matching answers say little about a pair; low bars are the questions that actually separate students. The most divisive are E02 (written exams measure knowledge, $E_q = 0.58$), T08 (AI diagnosis, 0.58), E04 (online learning, 0.63) and T09 (autonomous vehicles, 0.64); the most consensual are E15 (1.55), V13 (1.52), S05, E11 and V09 (1.43).

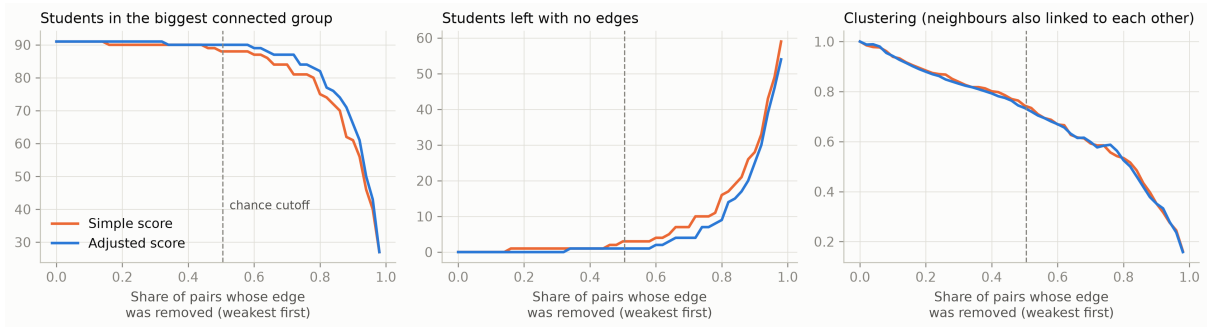


Figure 3: Removing the weakest edges step by step, comparing the simple and adjusted scores at equal density. The biggest connected group erodes gradually — there is no percolation-style collapse. The first students cut off are 120 (no above-chance edge at all), 89, 63 and 22, most of whom answer well below the class average of +1.14. Beyond about 90% removed the curves describe an empty network rather than the class.

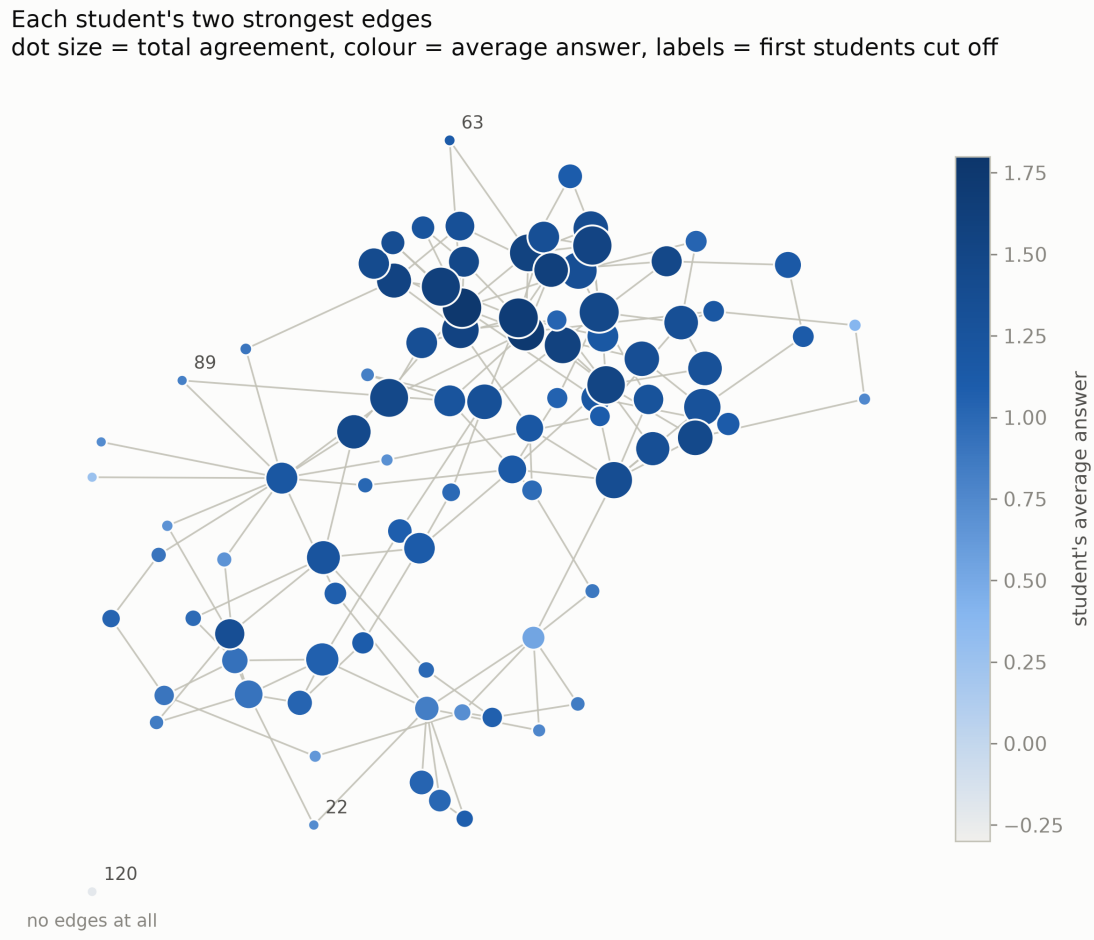


Figure 4: The network reduced to each student's two strongest edges (155 edges, 3.8% of pairs). 90 of the 91 students remain in one piece. Dot size is total agreement (strength) in the full network and colour is the student's average answer: the dark, emphatic answerers form the dense upper core, lighter and more hesitant students sit on the rim, and student 120 has no edge at all.

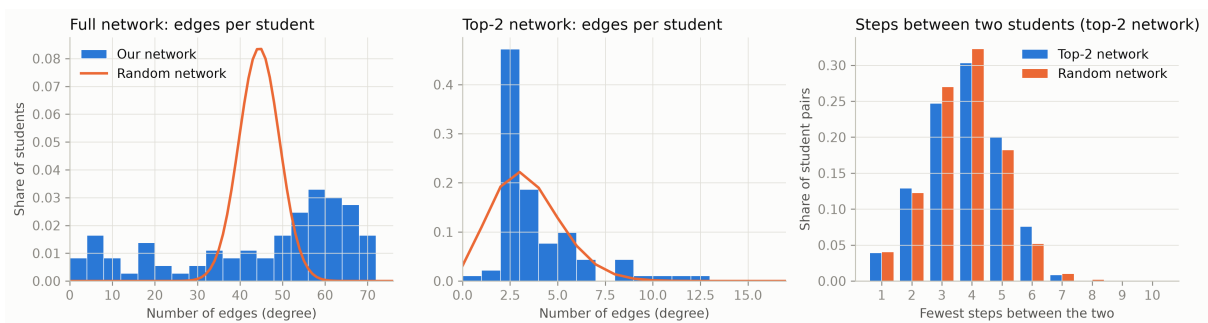


Figure 5: Left and centre: number of edges per student in the full and top-2 networks, against the binomial curve of an ER graph of the same size. Both are far broader than random. Right: distances in the top-2 network are essentially the random values, so the network is short-pathed without being random.

	Full network	ER graph	Top-2 network	ER graph
Edges	2,026	2,026	155	155
Density	0.495	0.495	0.038	0.038
Mean degree	44.53	44.53	3.41	3.41
Degree spread (σ_k)	21.00	4.59	2.23	1.80
Average clustering	0.730	0.494	0.154	0.026
Transitivity	0.757	0.494	0.090	—
Average shortest path	1.52	1.51	3.76	3.65
Diameter	4	2	7	≈ 5
Global efficiency	0.732	0.747	0.308	0.338
Largest component	90	91	90	87.6
Small-world σ	1.46	—	5.66	—
Degree assortativity	0.027	−0.023	−0.323	−0.023

Table 1: Network A against ER graphs with the same number of nodes and edges (each ER column is the mean over 20 such graphs).

Reading the measurements. Clustering is far above random in both versions — six times random in the top-2 network, where each student contributed only two edges — while distances match the random values. That is the small-world signature, $\sigma = 1.46$ in the full network and 5.66 in the sparse one. Degrees are much more spread out than an ER graph allows (21.0 against 4.6), but bounded: with 91 students there is no evidence of a power-law. The negative assortativity of the top-2 network is largely constructed, since students naming the same popular classmate leave that classmate ringed by low-degree neighbours.

Five centrality measures are used, each capturing a different sense of being well placed in the network. They are defined here once and reused for the question networks in Sections 5.3 and 5.4.

Measure	Definition	Meaning in Network A
Degree	k_i , the number of edges, divided by $N - 1$	how many classmates this student agrees with more than chance
Strength	$s_i = \sum_j w_{ij}$	the same, weighted by how far above chance each edge is
Closeness	$(N - 1)$ divided by the total distance from i to everyone else	how short the chains of agreement from this student to the rest are
Betweenness	$\sum_{s \neq i \neq t} \sigma_{st}(i) / \sigma_{st}$, the share of shortest paths through i	whether the student sits between others, like a bridge
Eigenvector	$x_i = \lambda^{-1} \sum_j w_{ij} x_j$	agreeing with classmates who are themselves central

Path-based measures come in two versions: unweighted, where every edge is one step, and weighted, where an edge of weight w_{ij} has length $1/w_{ij}$, so strong agreement is a short step. Student 120 has no edges, so eigenvector centrality is computed on the connected part of the network.

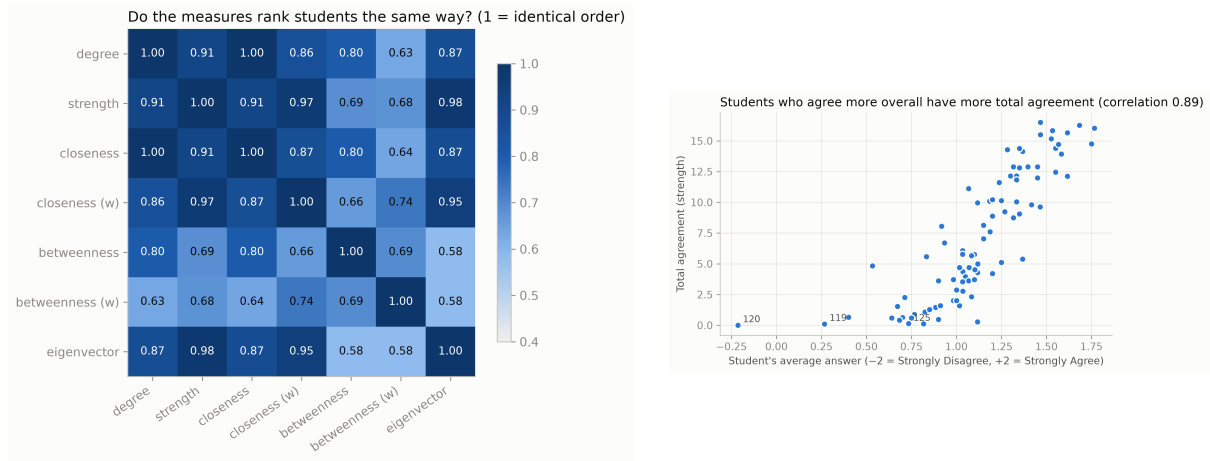


Figure 6: Left: how similarly the centrality measures rank the 91 students. Degree, closeness, strength and eigenvector agree closely ($\rho \approx 0.85$ – 1.00); betweenness is the odd one out ($\rho = 0.58$ – 0.80). Right: a student's total agreement tracks their average answer almost perfectly ($\rho = 0.89$).

Two kinds of central student. Strength and eigenvector centrality favour the emphatic answers (67, 110, 41, 114, 105), whose edges are the heaviest and mostly with each other. Degree and closeness favour a different set (52, 99, 101, 38, 53) who mix Agree and Strongly Agree and have the *most* edges. The same emphatic core reappears in the top-2 network, so it is not an artefact of density. The periphery is stable across every measure: 120, 125, 119, 89, 22, 63. Betweenness is led by 87 and 77, who answered only 15 questions, so that ranking is unreliable rather than meaningful.

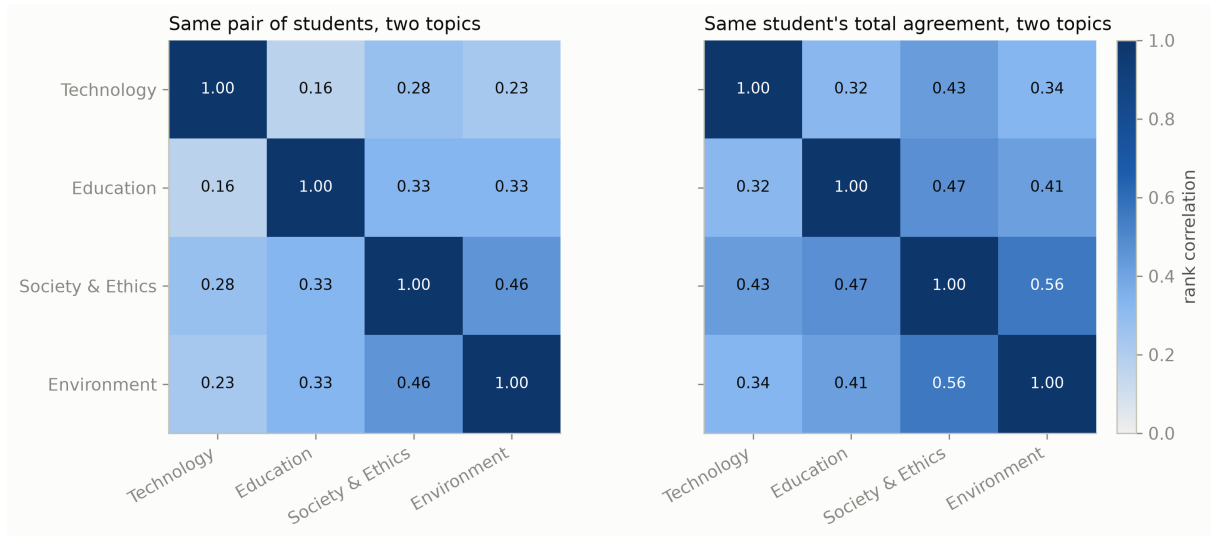


Figure 7: One network per topic. Left: how similarly two topics score the *same pair* of students. Right: how similarly they rank the same student's total agreement. Society and Environment line students up most alike (0.46 and 0.56); Technology and Education least (0.16). Nothing approaches 1.

Each topic alone produces a network much like the whole survey (roughly half of pairs linked, clustering 0.67–0.75), but the correlations above are all well below 1: **who a student agrees with depends on the topic**, and agreement is not one single attitude.

5.2 Network B: student-question extremes

Code and figures: *Extreme Bipartite Network/main.ipynb* (Tanush Garg). Every value quoted below was re-derived from the raw CSV for this report.

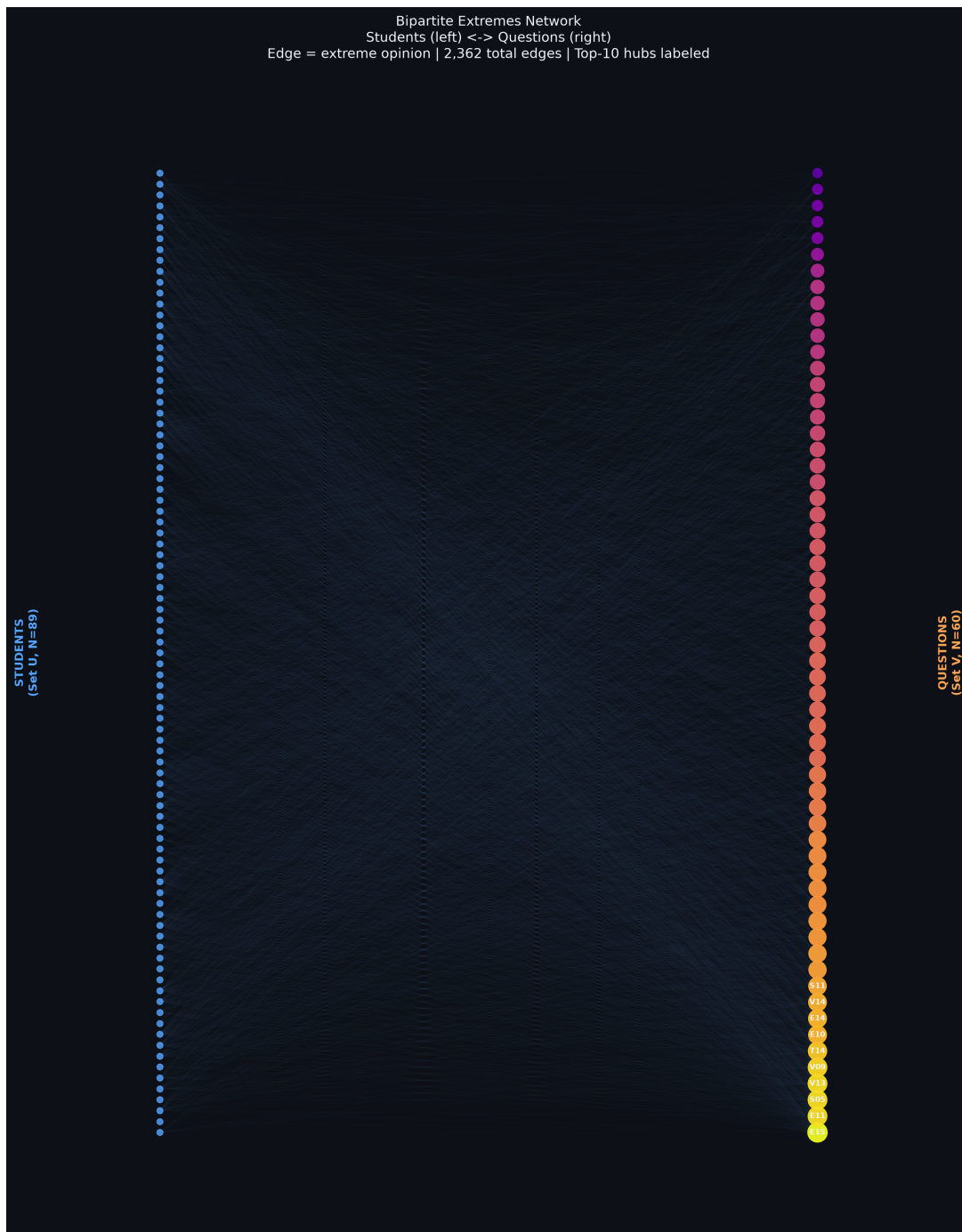


Figure 8: The bipartite extremes network: students (left) joined to the questions (right) on which they hold a Strongly Agree or Strongly Disagree view. Question nodes are coloured by degree; the ten largest “gravity wells” are labelled.

With 2,362 edges over a possible $89 \times 60 = 5,340$, the bipartite density is 0.4423: the cohort holds extreme views broadly rather than on a handful of pet topics. The filter retains 45.3% of the 5,218 valid responses, and the graph has exactly one connected component.

One property of this network deserves emphasis, because it governs how its results should be read: **2,251 of the 2,362 edges (95.3%) are Strongly Agree and only 111 are Strongly Disagree.** The “extremes” network is therefore very nearly a strong-*agreement* network. That explains why its anchors (E15, E11, V13) are the same statements Network A identifies as the most consensual: an edge here records intensity of endorsement, not opposition, so questions approaching unanimity dominate. Section by section, the mean question degree is highest for Environment (44.5) and Society (41.6) and *lowest* for Technology (33.9), so the extreme opinion in this cohort is concentrated in environmental and civic statements rather than technological ones — the mirror image of Network A’s finding that the most divisive questions are largely technological.

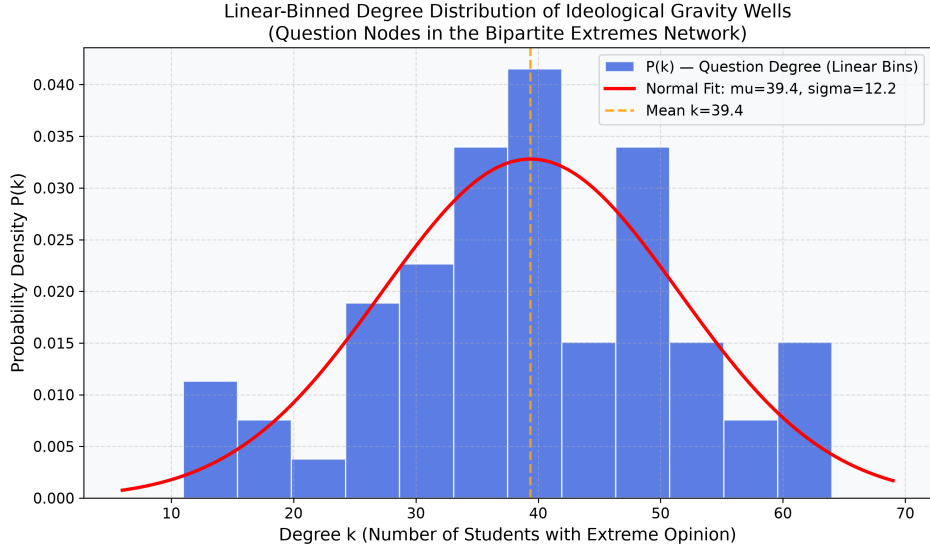


Figure 9: Linear-binned degree distribution of the 60 question nodes with a fitted normal curve ($\mu = 39.37$, $\sigma = 12.16$; range 11–64). The peaked, bounded shape rules out the monotonic decay of a power law.

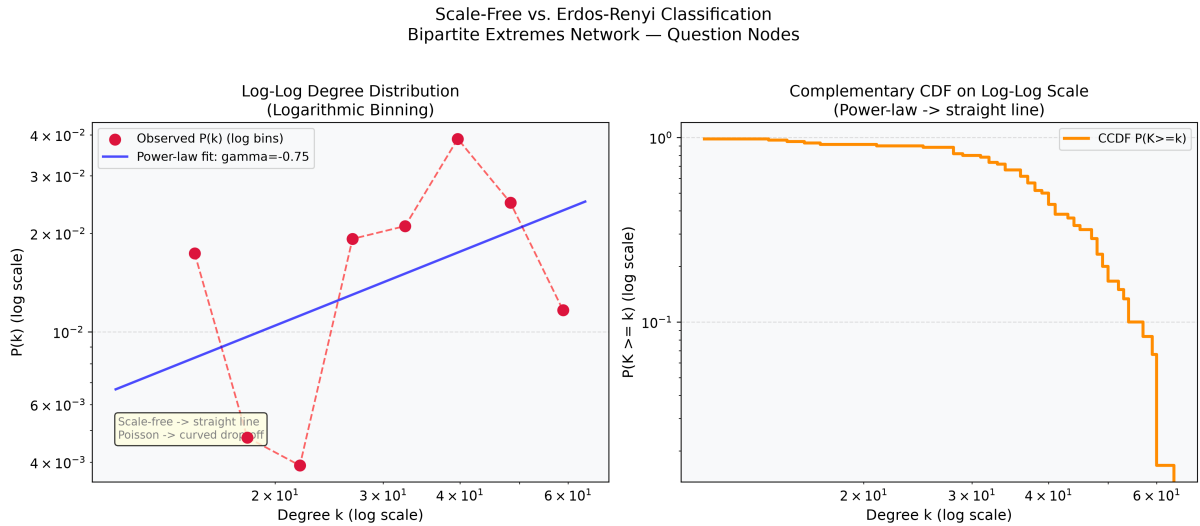


Figure 10: Log-log degree distribution with logarithmic binning (left) and the complementary CDF (right). The best-fit exponent $\hat{\gamma} = 0.750$ lies far outside the $2 < \gamma < 3$ band required of a scale-free network, and both curves bend rather than run straight.

The distribution is not Poisson either: the dispersion index $\text{Var}(k)/\langle k \rangle = 147.9/39.4 \approx 3.75$ is nearly four times the ER expectation of 1. The extremes network is **bounded but overdispersed** — no mega-hubs, yet some questions attract far more intensity than chance allows.

5.3 Network C: question–question projection

Code and figures: *Extreme Bipartite Network/main.ipynb* (Tanush Garg).

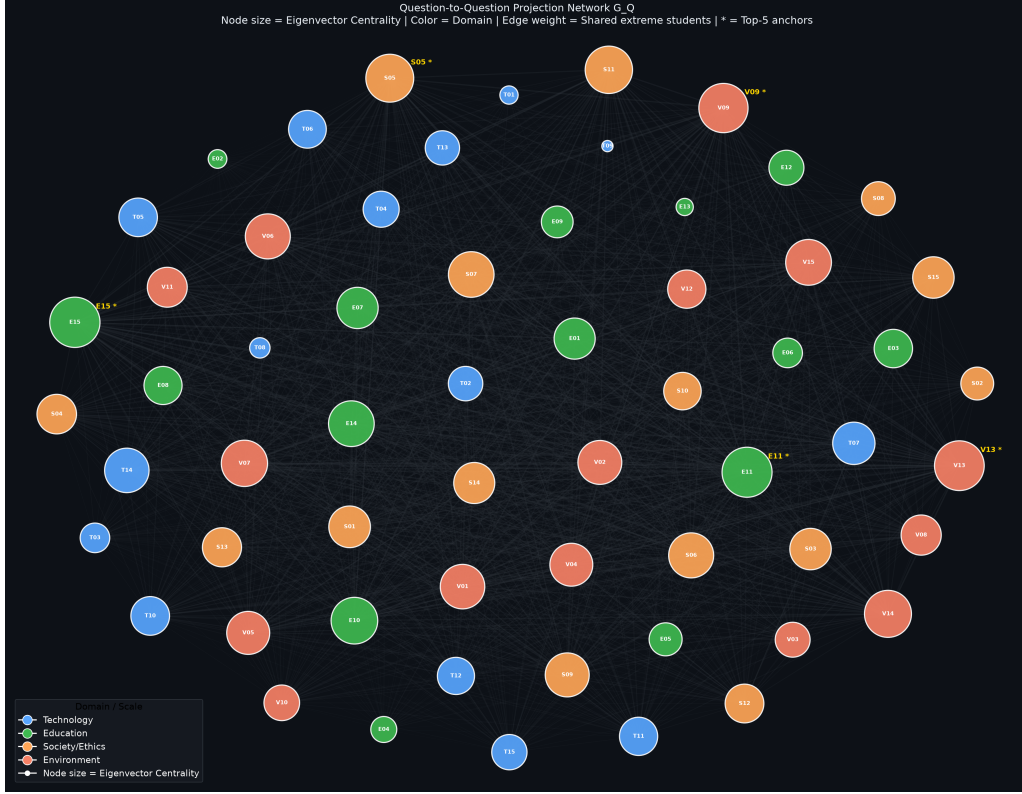


Figure 11: The question–question projection. Every pair of questions is joined, i.e. the unweighted projection is the complete graph K_{60} (1,770 edges, density 1.0, clustering 1.0, average path length 1.0). Colour marks the four survey domains.

Completeness is itself the finding: **for every pair of questions there is at least one student holding extreme views on both**, so extreme opinion in this cohort is not organised into subject-specific silos. It also means unweighted centrality is uniform — every question has degree 59 — which is why the weighted projection G_Q^W carries the analysis. Three measures are used on it, the same ones defined in Section 5.1, now applied to questions: **strength** $C_S(v) = \sum_u w_{vu}$, the total number of shared extreme views, normalised by the largest such total; **weighted betweenness**, computed with edge lengths $d_{ij} = 1/w_{ij}$ so that heavily shared pairs count as close together; and **weighted eigenvector** centrality, $C_E(v) = \lambda^{-1} \sum_u w_{vu} C_E(u)$, which rewards questions co-held with other central questions.

Centrality Analysis: Identifying Ideological Gravity Wells
(Top 15 Questions in the Weighted Projected Network G_Q^W)

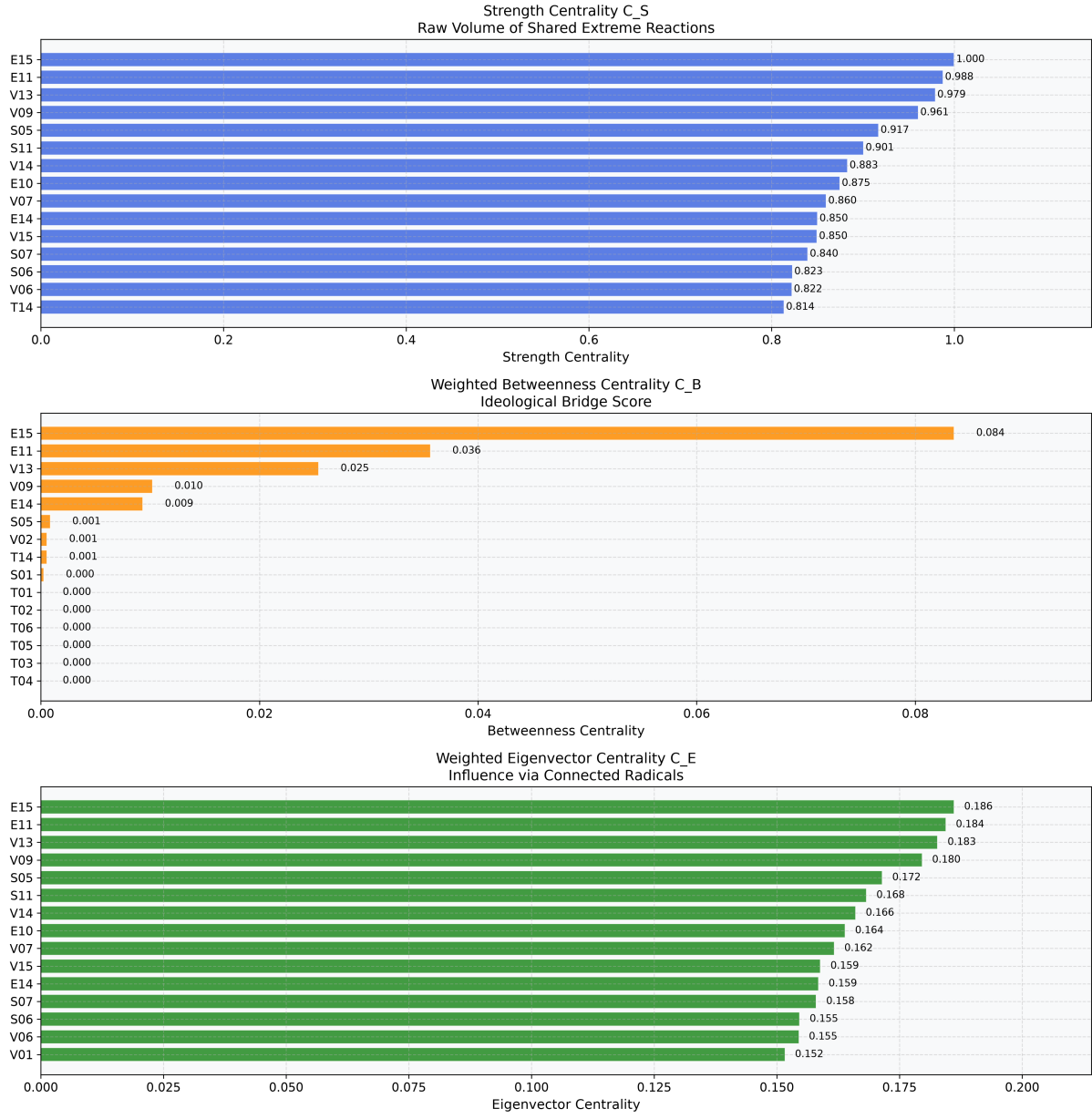


Figure 12: Strength, weighted betweenness and weighted eigenvector centrality on G_Q^W . Strength and eigenvector decay gradually; betweenness is sharply skewed, collapsing after the top five questions.

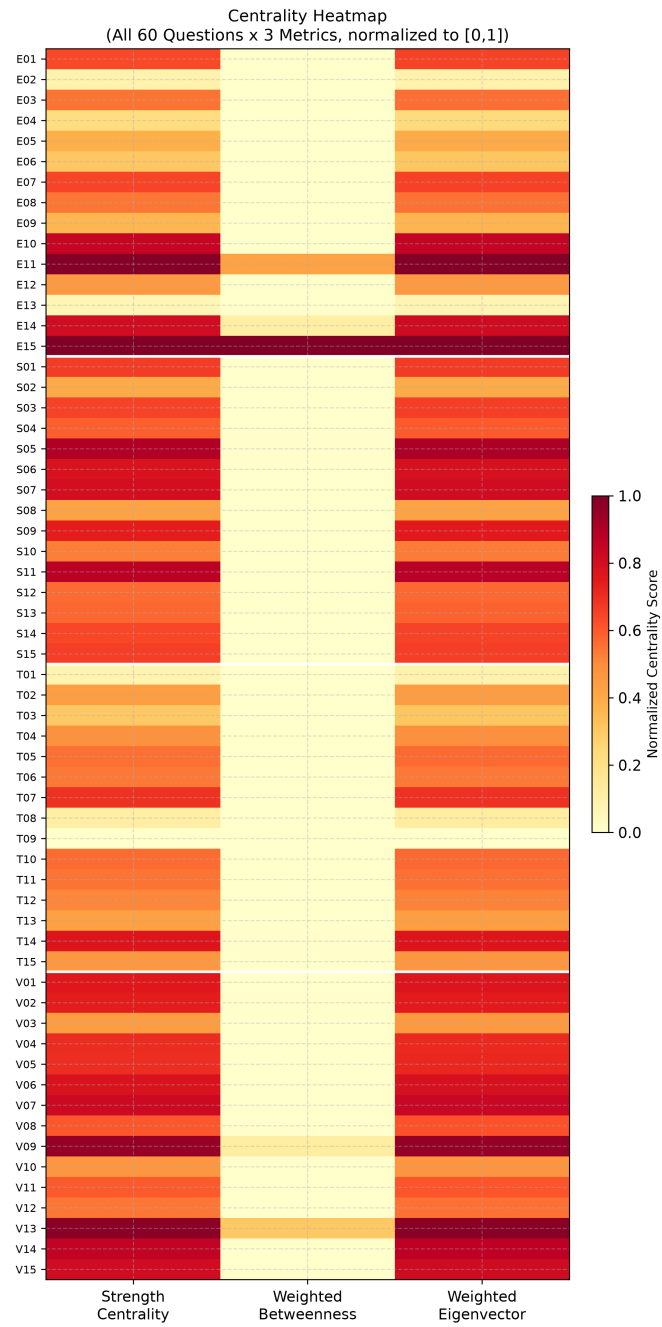


Figure 13: All 60 questions across the three weighted measures. The hot rows are the questions that act as ideological anchors.

ID	Statement	Strength	Eigenvector
E15	Continuous learning and skill development are essential throughout one's career	1.000	0.186
E11	University curricula should be updated more frequently for emerging technologies	0.988	0.184
V13	Companies should be accountable for their environmental impacts	0.979	0.183
V09	Products should be designed for reuse, repair and recycling	0.961	0.180
S05	Individuals should control how their personal data are collected and used	0.917	0.172
S11	People should be free to express differing opinions (absent harm)	0.901	0.168
V14	International cooperation is necessary for environmental challenges	0.883	0.166
E10	Universities should prioritise innovation over rote learning	0.875	0.164

Table 2: The strongest ideological anchors in G_Q^W . Weighted betweenness ranks the same three questions first (E15 0.084, E11 0.036, V13 0.025) before collapsing towards zero.

The **Molloy–Reed criterion** gives the condition for a giant component to exist in a network with a given degree sequence: $\kappa = \langle k^2 \rangle / \langle k \rangle > 2$, where $\langle k \rangle$ is the mean degree and $\langle k^2 \rangle$ the mean squared degree. Here every question has degree 59, so $\langle k \rangle = 59$, $\langle k^2 \rangle = 3,481$ and $\kappa = 59$ — the largest value a 60-node graph can reach, and far above the threshold of 2.

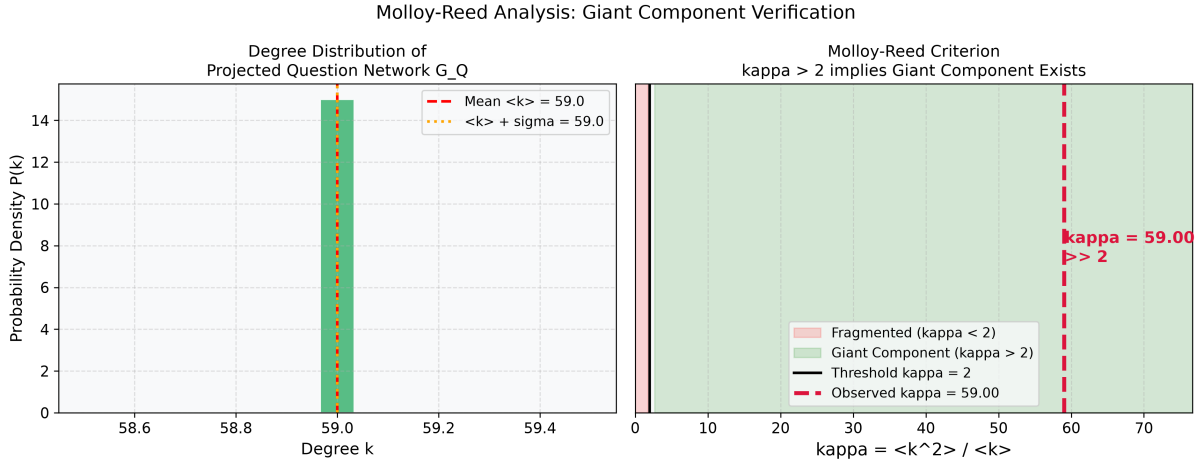


Figure 14: Molloy–Reed criterion on the unweighted projection. Every node has degree 59, giving $\kappa = \langle k^2 \rangle / \langle k \rangle = 59 \gg 2$: the giant component is guaranteed, by the widest possible margin for a 60-node graph.

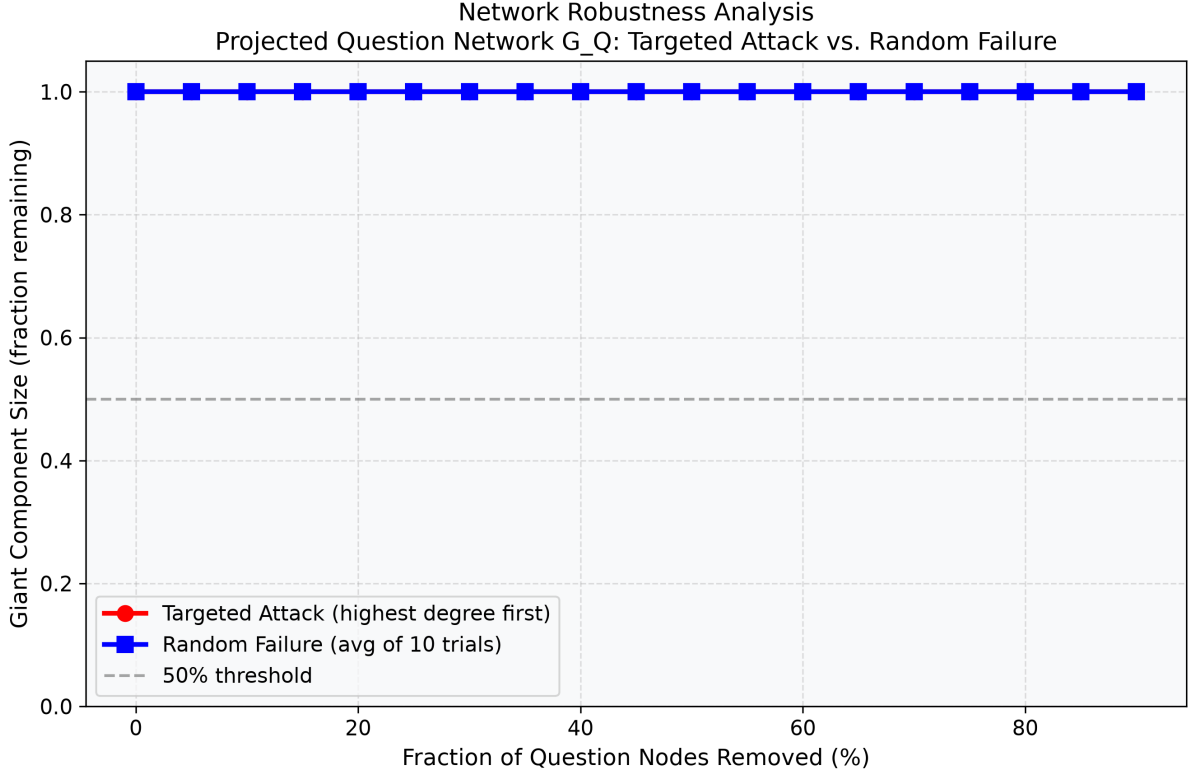


Figure 15: Robustness of the unweighted projection under targeted attack (highest degree first) and random failure. In K_{60} the two are mathematically indistinguishable and the giant component holds at 1.0 until the graph is nearly exhausted, because removing a node from a complete graph leaves a complete graph.

5.4 Network D: question-question correlation

Code and figures: `Question Network/question_network.ipynb` (Amish Goyal). Every value quoted below was re-derived from the raw CSV for this report.

At the $\rho > 0.35$ cutoff the network keeps **188 of the 1,770 possible edges** (density 0.106, average degree 6.27). It is not one connected whole: 47 questions form a giant component, and **13 questions are left isolated**, correlating strongly with nothing at all. This is the first of the four networks that fragments, and the fragmentation is the finding.

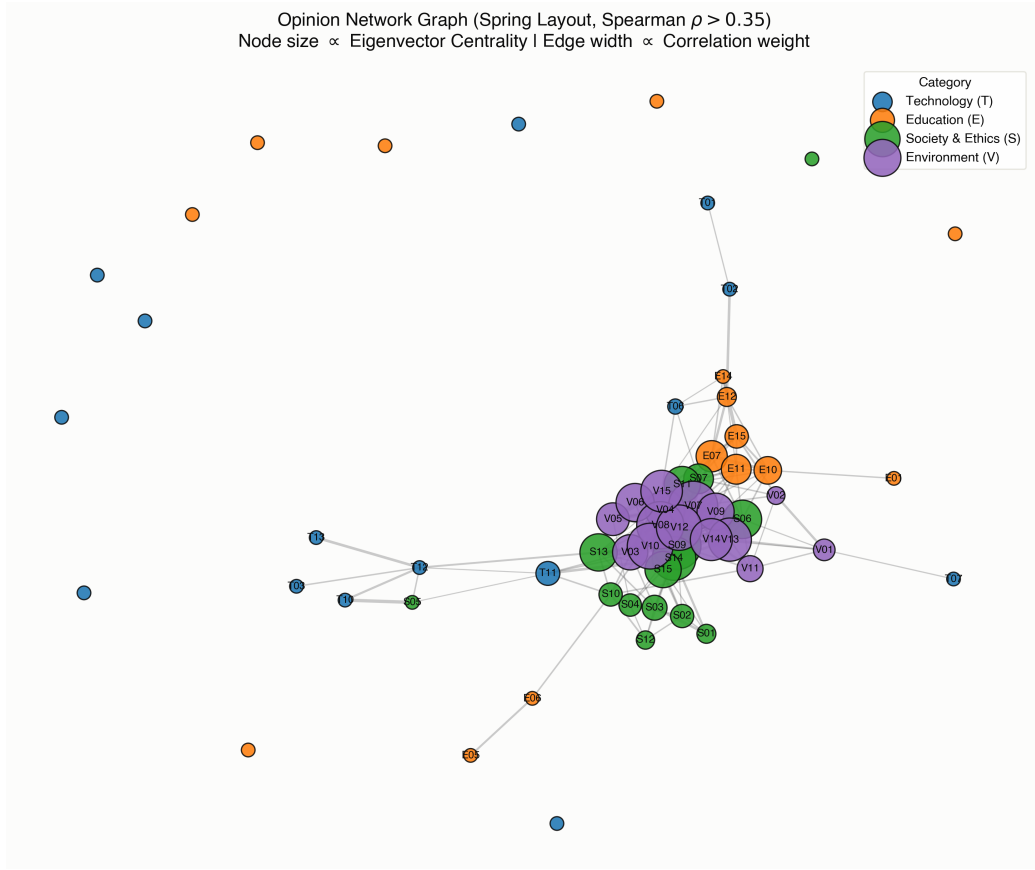


Figure 16: The question-question correlation network. Node colour marks the survey section (blue Technology, orange Education, green Society & Ethics, purple Environment), node size is eigenvector centrality and edge width is ρ . Sections mix, but the environmental block stands out as a dense, almost unmixed region.

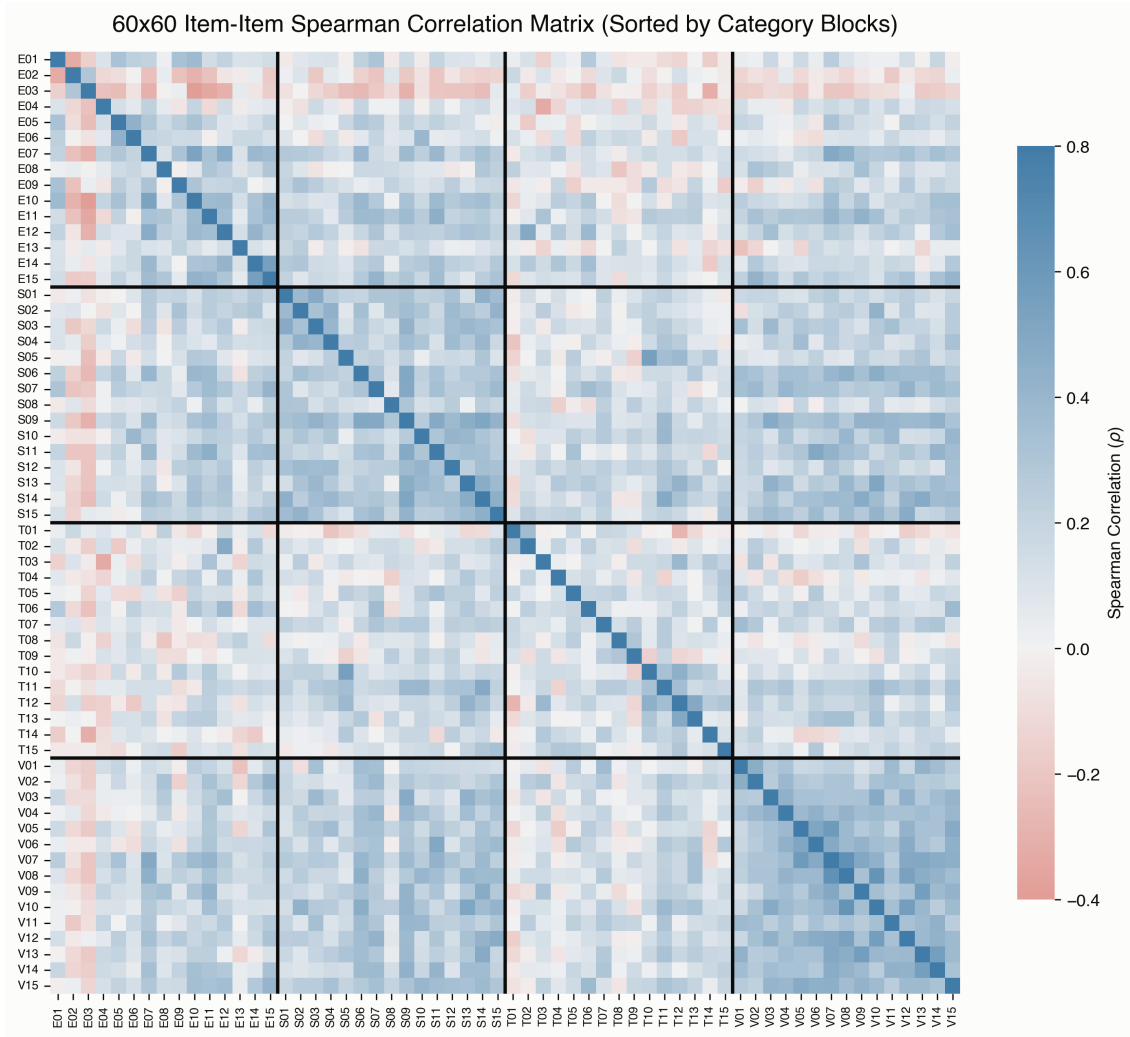


Figure 17: All 60×60 Spearman correlations, with the statements grouped by section (in the order E, S, T, V). The strongest block by far is Environment, in the lower right: environmental statements move together. The Technology block is visibly paler, so technology opinions co-vary far less. The two red rows near the top are E02 (written exams measure knowledge) and E03 (compulsory attendance), which correlate *negatively* with much of the survey and therefore join nothing at a positive cutoff.

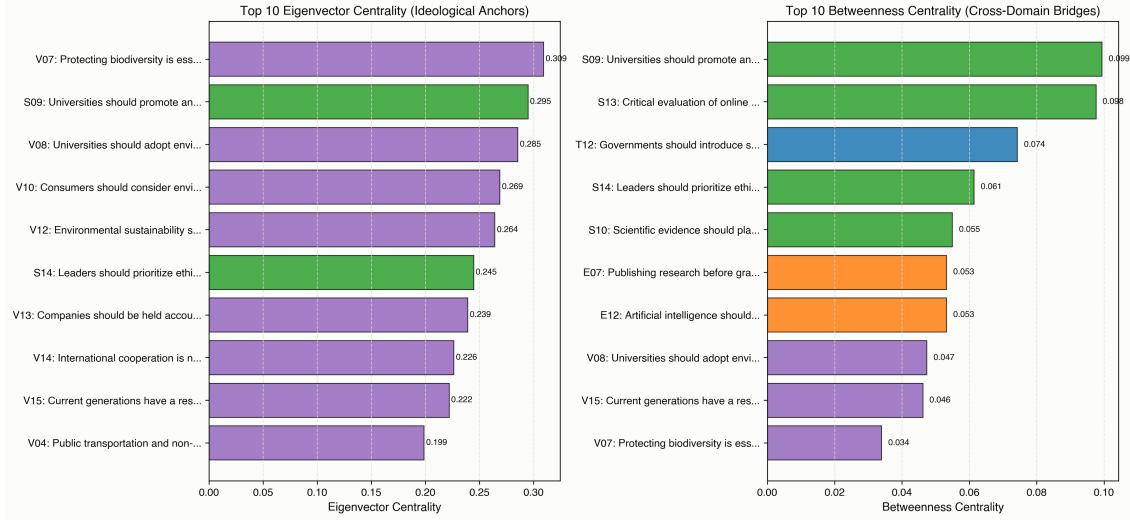


Figure 18: Top ten statements by eigenvector centrality (ideological anchors, left) and by weighted betweenness (cross-domain bridges, right).

Role	ID	Section	Score	Statement
Anchor (eigenvector)	V07	V	0.309	Protecting biodiversity is essential for long-term well-being
	S09	S	0.295	Universities should promote an inclusive environment
	V08	V	0.285	Universities should adopt sustainable campus practices
	V10	V	0.269	Consumers should consider environmental impact when purchasing
	V12	V	0.264	Sustainability should be integrated into higher education
Bridge (betweenness)	S09	S	0.099	Universities should promote an inclusive environment
	S13	S	0.098	Critical evaluation of online information should be taught
	T12	T	0.074	Governments should introduce stricter AI regulation
	S14	S	0.061	Leaders should prioritise ethical decision-making
	S10	S	0.055	Scientific evidence should inform public policy

Table 3: Anchors and bridges in Network D. Eight of the top ten anchors are environmental statements.

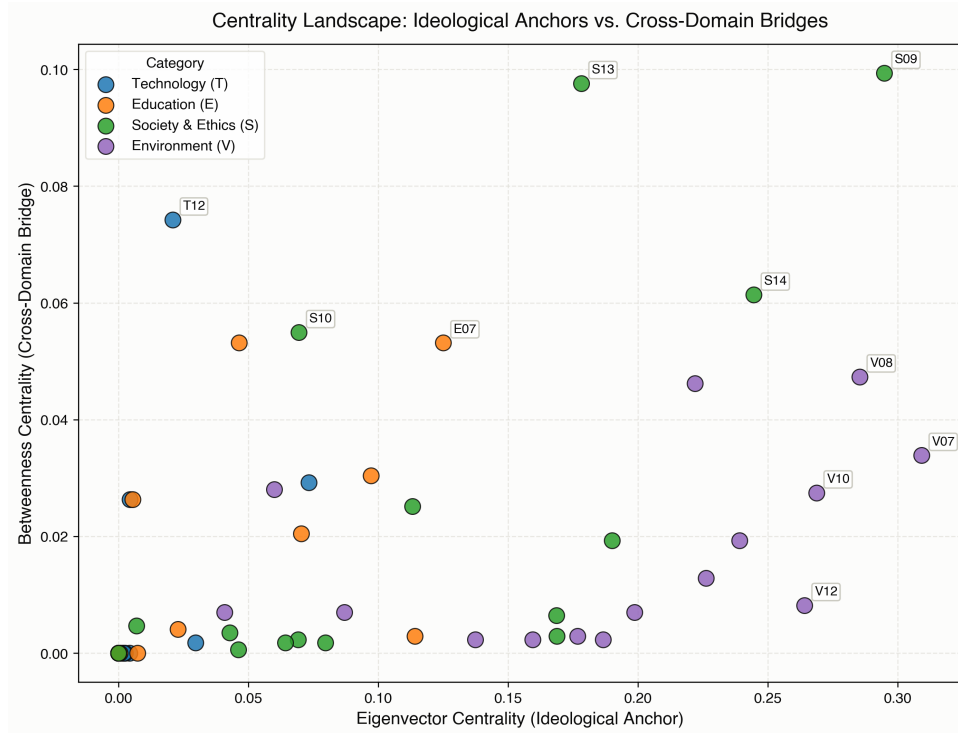


Figure 19: Eigenvector against betweenness centrality. Most statements sit near the origin; S09 is the rare statement that is both an anchor and the network’s primary bridge, while S13 and T12 bridge without being anchors.

Reading Network D. Three results stand out.

- **The belief core is environmental and ethical.** Eight of the ten highest eigenvector scores are Environment statements, interlocked with the ethics items S09 and S14. Agreement on biodiversity, sustainable campuses and responsible consumption moves as one block.
- **The bridges are the contested interfaces.** S09 (inclusion), S13 (digital literacy) and T12 (AI regulation) sit between clusters. They are where technology opinion connects to educational and civic opinion, and therefore where a change of mind would propagate furthest.
- **Beliefs do not follow the syllabus.** Louvain finds six organic modules, and section assortativity is only $r = 0.304$: sections matter, but students do not compartmentalise by them. Two modules are genuinely mixed — AI governance (T03, T10, T12, T13 with S05) and educational modernisation (education items with T01, T02, T06, S07, S11, V02) — while ecology forms an almost purely environmental module.

6 Results and Discussion

1. The class agrees far more than it disagrees, and that shapes every network. Four in five answers are on the agree side. In Network A this makes raw agreement useless as a similarity measure — it links everyone to everyone — and forces the chance correction. In Network B the same fact appears as a bipartite density of 0.44, and in Network C as a complete projection. Only Network D, which measures co-movement rather than agreement, escapes it: at $\rho > 0.35$ just 10.6% of question pairs survive.

2. Intensity and alignment pick out different questions, and the contrast is the most informative result in the project. Network A finds E15, V13, S05, E11 and V09 to

be the *most consensual* questions, the ones where a matching answer says almost nothing about a pair. Network C finds exactly those questions to be the strongest ideological anchors, because a near-unanimous Strongly Agree makes a question both the biggest gravity well for extreme opinion and the least informative for telling students apart. Network D then ranks that same E15 **26th** of the 47 questions in its giant component: with almost no variance left, a question everyone endorses cannot correlate with anything. Its anchors are instead V07, S09, V08, V10 and V12 — statements that people genuinely vary on, and vary on *together*. Read as a set: E15 is what the class believes, V07 and S09 are what its beliefs are organised around.

3. The questions that split the class are the ones that belong to no structure at all. Network A ranks statements by chance agreement E_q ; its five most divisive are E02 (written exams measure knowledge, $E_q = 0.58$), T08 (AI diagnosis, 0.58), E04 (online learning, 0.63), T09 (autonomous vehicles, 0.64) and E13 (research over infrastructure, 0.72). Network D, built by an entirely different route, leaves 13 questions isolated at $\rho > 0.35$: E02, E03, E04, E08, E09, E13, S08, T04, T05, T08, T09, T14 and T15. **All five of Network A’s most divisive statements are in that list.** One method compares pairs of students, the other pairs of questions, and they agree: where this class divides, it divides idiosyncratically, without lining up along any shared axis.

4. Nothing fragments except where we force it to. Network A’s largest component erodes one student at a time under thresholding, with no percolation-style collapse; Network B has a single connected component; Network C is complete, with $\kappa = 59 \gg 2$ and no distinction between random failure and targeted attack. Network D fragments only because a cutoff is imposed, and what falls out — the 13 isolates — is precisely the contested material. The cohort has no echo chambers and no topic silos.

5. Structure is small-world and broad, but never scale-free. Network A is clustered well above random ($6\times$ in its sparse version) at random-like distances, $\sigma = 1.46$ and 5.66 . Network B’s question degrees are overdispersed (dispersion index 3.75) but bounded, with $\hat{\gamma} = 0.750$, nowhere near the scale-free band. Both are broader than an ER graph and far from a power law — unsurprising at $N = 91$ and $N = 60$, and worth stating as a limit rather than a discovery.

6. Centrality in Network A largely measures agreeableness; in Network D it measures alignment. A student’s total agreement tracks their average answer at $\rho = 0.89$, and the emphatic answerers hold the heaviest edges, so being “central” among students mostly means agreeing readily. Among questions, centrality means something sharper: S09 is simultaneously an anchor and the primary bridge, so inclusion is both a core commitment and the hinge connecting educational, civic and environmental opinion.

7. Topic matters, and the two halves of the project agree on how. Network A finds Society and Environment line students up most alike (pair scores 0.46, strengths 0.56) and Technology and Education least (0.16). Network D finds the same asymmetry from the question side: ecology forms an almost unmixed module and the environmental block is the brightest region of the correlation heatmap, while technology items scatter into AI-governance and educational-modernisation modules. Section assortativity of $r = 0.304$ says the survey’s own four sections are a weak guide to how these students actually think.

Limitations. The cohort is small (91 usable responses), so degree-distribution claims are bounded by N ; three respondents in Networks B and C answered only 15 of the 60 questions, which understates their degree there and inflates their betweenness in Network A; Network D’s structure depends on one free parameter, the $\rho > 0.35$ cutoff, which sets how many questions end up isolated; and a Likert survey cannot separate genuine conviction from a habit of reaching for the end of a scale, so “emphatic” should be read as a response pattern, not a personality.

7 Individual Contribution

Member	Contribution
Mayank Kar	Network A (student–student), and this combined report. Shared cleaning and encoding module <code>clean.py</code> used by the whole team; the simple and chance-adjusted agreement scores; the sparsification experiments (global threshold sweep and the top-2 construction); the structural measurements against ER graphs, including clustering, path lengths, efficiency, small-world σ and assortativity; all five centrality measures and their rank comparison; the per-topic networks; <code>student_network.ipynb</code> and the eight figures in <code>student_plots/</code> .
Tanush Garg	Networks B and C (student–question bipartite and its question–question projection). The extremes filter and long-format edge list; construction and validation of the bipartite graph; bipartite topology (density, components, isolated respondents); question-node degree distribution with linear and logarithmic binning, the power-law fit and the dispersion analysis; the projection onto the question set and the weighted projection G_Q^W ; strength, weighted betweenness and weighted eigenvector centrality on it; the Molloy–Reed evaluation and the robustness simulation; <code>main.ipynb</code> , the eight figures in <code>extreme_plots/</code> and the interactive D3 visualisers.
Amish Goyal	Network D (question–question correlation). Pairwise Spearman correlation across all 1,770 question pairs on the shared cleaned data; the thresholded graph at $\rho > 0.35$ with affinity weights and inverse- ρ distances; weighted eigenvector and metric betweenness centrality; Louvain community detection and the interpretation of the six modules; the section assortativity coefficient and identification of the isolated questions; <code>question_network.ipynb</code> , the four figures in <code>Question Network/</code> , the repository’s initial setup and its later reorganisation into per-network folders, and the project landing page <code>index.html</code> .